7.13 Multiple Timestepping

Recall the r-RESPA multiple timestepping scheme introduced in Sect. 4.2, designed to propagate systems which have a fast highly oscillatory motion, as well as a slower long-ranged force term. Splitting the potential energy function into fast and slow components in the deterministic setting provided only a small gain in the stability threshold, as we encountered linear resonance phenomena at step sizes that were integer multiples of half a period. In systems where the timescale gap is very large, the r-RESPA scheme does not give significant improvement, with the stability threshold scaling identically to the Symplectic Euler method.

Our hope would be that using Langevin dynamics to add stochasticity would disrupt resonance effects and permit larger usable timesteps. While some considerable efficiency improvements have been demonstrated using multiple timestepping methods for Langevin dynamics with constraints [118], we shall investigate the simpler fast/slow splitting demonstrated in previous chapters.

We consider propagating a system with Hamiltonian

$$H(q,p) = p^2/2 + U(q), \qquad U(q) = U_{\mathrm{F}}(q) + U_{\mathrm{S}}(q),$$

where $q, p \in \mathbb{R}$ and U_{S} and U_{F} denote the slow and fast components of the potential energy respectively. For the purposes of analysis, we shall model these with harmonic potentials, with

$$U_{\mathrm{S}}(q) = q^2/2, \qquad U_{\mathrm{F}}(q) = \Omega^2 q^2/2, \tag{7.47}$$

where $\Omega \gg 1$. The question we shall be interested in is whether using Langevin dynamics with a multiple timestepping algorithm gives us any extra stability compared to the deterministic case. Let us consider splitting the dynamics into

$$\begin{bmatrix} \mathrm{d}q \\ \mathrm{d}p \end{bmatrix} = \underbrace{\begin{bmatrix} p\,\mathrm{d}t \\ -\Omega^2 q\,\mathrm{d}t - \gamma p\,\mathrm{d}t + \sqrt{2\gamma/\beta}\,\mathrm{dW} \end{bmatrix}}_{\text{Fast}} + \underbrace{\begin{bmatrix} 0 \\ -q\,\mathrm{d}t \end{bmatrix}}_{\text{Slow}}, \tag{7.48}$$

and proceed as in the case of the r-RESPA scheme. As before, we consider the case where we are able to solve the fast part exactly (in practice we use several small timesteps as the fast parts are considered computationally "cheap"). Writing

$$\boldsymbol{F} = \begin{bmatrix} 0 & 1 \\ -\Omega^2 & -\gamma \end{bmatrix},$$

we can write down the solution to the fast dynamics as

$$\begin{bmatrix} q(t) \\ p(t) \end{bmatrix} = \exp(\boldsymbol{F}t) \begin{bmatrix} q(0) \\ p(0) \end{bmatrix} + \boldsymbol{B}_t \mathbf{R},$$

where $\mathbf{R}$ is a column vector of two i.i.d. random normal numbers distributed as $\mathcal{N}(0,1)$, and

$$\boldsymbol{B}_t\boldsymbol{B}_t^T = \frac{2\gamma}{\beta}\int_0^t \exp(\boldsymbol{F}(t-s))\begin{bmatrix}0&0\\0&1\end{bmatrix}\exp(\boldsymbol{F}^T(t-s))\mathrm{d}s.$$

The solution to the slow part of our splitting is just

$$\begin{bmatrix}q(t)\\p(t)\end{bmatrix} = \begin{bmatrix}1&0\\-t&1\end{bmatrix}\begin{bmatrix}q(0)\\p(0)\end{bmatrix}.$$

Thus for a method taking one step of the fast system followed by a step of the slow system, using fixed step size $h > 0$, we can write the update scheme for $\boldsymbol{z} = [q,p]^T$ as

$$\boldsymbol{z}_{n+1} = \boldsymbol{X}\boldsymbol{z}_n + \boldsymbol{Y}\mathbf{R}_n,$$

for

$$\boldsymbol{X} = \begin{bmatrix}1&0\\-h&1\end{bmatrix}\exp(h\boldsymbol{F}), \qquad \boldsymbol{Y} = \begin{bmatrix}1&0\\-h&1\end{bmatrix}\boldsymbol{B}_h.$$

The stability of the method therefore depends solely on the spectral radius of the matrix $\boldsymbol{X}$. Denoting $\eta = \sqrt{\gamma^2 - 4\Omega^2}$, then we compute

$$\det(\boldsymbol{X}) = e^{-\gamma h}, \qquad \mathrm{tr}(\boldsymbol{X}) = \frac{\eta - h}{\eta}e^{(\eta-\gamma)h/2} + \frac{\eta+h}{\eta}e^{(-\eta-\gamma)h/2},$$

giving a stability condition

$$\left|\frac{\eta - h}{\eta}e^{(\eta-\gamma)h/2} + \frac{\eta+h}{\eta}e^{(-\eta-\gamma)h/2}\right| \leq 1 + e^{-\gamma h}.$$

Multiplying by $e^{\gamma h/2}$ gives

$$\left|\cosh\left(\frac{\eta h}{2}\right) + \frac{h}{\eta}\sinh\left(-\frac{\eta h}{2}\right)\right| \leq \cosh\left(\frac{\gamma h}{2}\right),$$

which reduces to the deterministic stability condition as expected in the limit $\gamma = 0$. For $\gamma > 0$, we can simplify to

$$\left|\cosh\left(\frac{\eta h}{2} - \mathrm{atanh}\left(\frac{h}{\eta}\right)\right)\sqrt{1 - \frac{h^2}{\eta^2}}\right| \leq \cosh\left(\frac{\gamma h}{2}\right).$$